Junhuai Xu

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Curriculum Vitae
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Profile

Experimental nuclear and particle physicist working at the intersection of detector instrumentation, scientific computing, and physics-informed reconstruction. My research spans short-range correlations in nuclei, atmospheric-neutrino simulations for deep-sea neutrino telescopes, and correlation-function imaging in heavy-ion collisions. I develop analysis workflows that connect detector calibration, response simulation, event reconstruction, inverse problems, statistical inference, and machine-learning methods.

Research Experience

Nucleon Short-Range Correlations — Oct. 2023 – Present
CSHINE Experiment, RIBLL-1 (HIRFL), Lanzhou, China

- Developed calibration and detector-response models for the CsI(Tl) γ-ray hodoscope used in hard-γ measurements from low-energy heavy-ion collisions.
- Implemented Richardson–Lucy deconvolution of bremsstrahlung spectra and quantified reconstruction and systematic uncertainties through pseudo-experiments and closure tests.
- Constrained IBUU-MDI transport calculations using high-statistics γ spectra of ^{124}Sn+^{124}Sn at 25 MeV/u, extracting an SRC-induced high-momentum-tail fraction of $(20 \pm 3)\%$ in ^{124}Sn nuclei.

Correlation Function Imaging — Feb. 2024 – Present
Tsinghua University, Department of Physics

- Developed a Richardson–Lucy imaging framework for femtoscopic two-particle correlations to reconstruct freeze-out source distributions beyond Gaussian parameterizations.
- Applied pp and $\bar{p}\bar{p}$ correlations in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV to infer non-Gaussian source functions and constrain strong-interaction parameters.
- Extended the approach toward three-dimensional source imaging with reproducible fit, resampling, and validation workflows for collaboration analyses.

Deep-Sea Neutrino Detection — Jun. 2024 – Present
TRIDENT Collaboration

- Built an atmospheric-neutrino simulation chain combining HONDA fluxes, FLUKA interactions, and Geant4 detector transport to generate labeled detector-response datasets for a compact TRIDENT array.
- Developed graph-neural-network reconstruction of event class, direction, and energy from simulated detector responses, and benchmarked performance for oscillation analyses.
- Performed oscillation-parameter sensitivity studies using Poisson-sampled pseudo-experiments and spectral comparisons across parameter hypotheses.

Experimental Experience

- **CSHINE Experiment** (Lanzhou, China, 2024): high-energy γ-ray measurements, detector response, and SRC studies in low-energy heavy-ion collisions.
- **SLEGS Beam Test at SSRF** (Shanghai, China, Jan. 2025): high-energy γ-ray response calibration of CsI(Tl) crystals using quasi-monochromatic photon beams and Geant4 simulations.
- **SπRIT Experiment** (RIKEN, Japan, 2024): supported detector operation and electronics control during SAMURAI beam time, contributing to stable data taking and run coordination.

Technical Skills

Physics analysis	Detector calibration, detector-response simulation, event reconstruction, spectrum unfolding, systematic uncertainty evaluation.
Inference	Inverse problems, Richardson–Lucy deconvolution/imaging, likelihood-based inference, pseudo-experiments, closure tests, resampling.
Machine learning	Graph neural networks, PyTorch, physics-informed reconstruction, simulation-to-inference workflows.
Computing	Python, C/C++, ROOT, Geant4, FLUKA-based workflows, Linux, Git, HPC/GPU-accelerated computing, LaTeX.

Education

Tsinghua University Aug. 2022 – Jun. 2027 (Expected)
Ph.D. Candidate in Physics
Research areas: short-range correlations, heavy-ion collisions, femtoscopy, neutrino simulation, and machine-learning-based reconstruction.

South China Normal University Aug. 2018 – Jun. 2022
Bachelor of Science
GPA: 4.06/5.00; Rank: 1/137. Thesis: *A Study of the Nuclear Medium Effect on Hadron Fragmentation Functions*.

Selected Publications

A complete publication list is available separately on my personal website.

- **Xu, J.** *et al.*, "Precise measurement of short-range correlations in nuclei from bremsstrahlung gamma-ray emission in low-energy heavy-ion collisions," ***Phys. Rev. Res.*** **7**, 043174 (2025).
- **Xu, J.** *et al.*, "Experimental study of bremsstrahlung γ-ray emission and short-range correlations in ^{124}Sn+^{124}Sn collisions at 25 MeV/nucleon," ***Phys. Rev. C*** **113**, 044613 (2026).
- **Xu, J.** *et al.*, "Reconstruction of Bremsstrahlung γ-rays spectrum in heavy ion reactions with Richardson-Lucy algorithm," ***Phys. Lett. B*** **857**, 139009 (2024).
- **Xu, J.** *et al.*, "Imaging Freeze-Out Sources and Extracting Strong Interaction Parameters in Relativistic Heavy-Ion Collisions," ***Chin. Phys. Lett.*** **42**, 031401 (2025).
- **Xu, J.** *et al.*, "Linear response of CsI(Tl) crystal to energetic photons below 20 MeV," ***Nucl. Instrum. Meth. A*** **1080**, 170787 (2025).
- Zhang, H., **Xu, J.** *et al.*, "Probing the three-dimensional emission source and neutron skin via π-π correlations in heavy-ion collisions," ***Phys. Rev. C*** **113**, 034904 (2026).

Awards and Selected Talk

- First-class Comprehensive Scholarship, Tsinghua University (2025).
- Second-class Comprehensive Scholarship, Tsinghua University (2024).
- Outstanding Teaching Assistant, Tsinghua University (2024).
- Presented a talk titled *Simulation of Atmospheric Neutrinos with a Compact Underwater Neutrino Telescope* at the 7th TRIDENT Collaboration Meeting, Sanya, Hainan, China (May 2026).
- First Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM, 2020); Third Prize, National Undergraduate Mathematics Competition Final Round, Non-Math Major (2021).